Introduction to Machine Learning for Text Analysis with Python
Day 5 – Friday
»Navigating the Model Ecosystem, Critical Evaluation, optional topics + Project Day 2«

Rupert Kiddle
Sjoerd Stolwijk

r.t.kiddle@vu.nl, @rptkiddle
s.b.stolwijk@uu.nl

September 18, 2026

Today

Recap

Model selection

Evaluation criteria for computational approaches

Method-immanent criteria

- **Accuracy** against a gold standard
- **Speed** of processing
- **Transparency** of the model's inner workings
- **Replicability** of classification results
- **Language coverage** — esp. non-English texts

Method-external criteria

- **Resources & competences** needed to apply the method
- **Ethical questions**, incl. external costs to nature & society:
  - energy consumption per task
  - data privacy / ownership rights

Source: Maurer et al., 2026

Showstoppers - Example

	Dictionary		SML	UML	DL	LLM	gLLM	
Show stopper:	off-the-shelf	custom-made					local	API
Data & task characteristics								
no (large) annotated dataset available	✓	✓	✗	✓	✗	✓	✓	✓
no language-specific tool available	✗	✓	✓	✓	✓	✗	✗	✗
categories known a-priori	✓	✓	✓	✗	✓	✓	✓	✓
categories not known a-priori	✗	✗	✗	✓	✗	✓	✓	✓
Resources & skills required								
no access to high-performance computing	✓	✓	✓	✓	(✓)	(✗)	(✗)	✓
no programming skills	(✓)	(✓)	✗	✗	✗	✗	✗	(✓)
no statistics skills	✓	(✗)	✗	✗	✗	✗	✓	✓

Showstoppers - Example

Show stopper:	Dictionary: off-the-shelf	Dictionary: custom-made	SML	UML	DL	LLM	gLLM: local	gLLM: API
Data & task characteristics								
no (large) annotated dataset available	✓	✓	✗	✓	✗	✓	✓	✓
no language-specific tool available	✗	✓	✓	✓	✓	✗	✗	✗
categories known a-priori	✓	✓	✓	✗	✓	✓	✓	✓
categories not known a-priori	✗	✗	✗	✓	✗	✓	✓	✓
Resources & skills required								
no access to high-performance computing	✓	✓	✓	✓	(✓)	(✗)	(✗)	✓
no programming skills	(✓)	(✓)	✗	✗	✗	✗	✗	(✓)
no statistics skills	✓	(✗)	✗	✗	✗	✗	✓	✓

Showstoppers - Example

	Dictionary		SML	UML	DL	LLM	gLLM	
Show stopper:	off-the-shelf	custom-made					local	API
Data & task characteristics								
no (large) annotated dataset available	✓	✓	✗	✓	✗	✓	✓	✓
no language-specific tool available	✗	✓	✓	✓	✓	✗	✗	✗
categories known a-priori	✓	✓	✓	✗	✓	✓	✓	✓
categories not known a-priori	✗	✗	✗	✓	✗	✓	✓	✓
Resources & skills required								
no access to high-performance computing	✓	✓	✓	✓	(✓)	(✗)	(✗)	✓
no programming skills	(✓)	(✓)	✗	✗	✗	✗	✗	(✓)
no statistics skills	✓	(✗)	✗	✗	✗	✗	✓	✓

Performance: How good is good enough

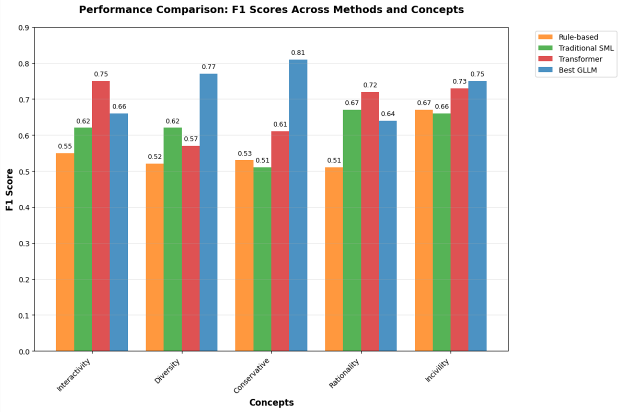


Figure 1: Overview of differences in macro average F1-macro scores for different classification approaches reported in Stolwijk et al., 2026

Pitfalls

Things to watch out for

- Label leakage
- Multiple comparisons: use your test set only once
- Old cached results - running notebook out of order
- Old cached results - not cleaning your model after a training run
- Typos and errors
- Quality of your manual baseline
- Good performance $\neq$ same result

The dangers of AI - yesterday's slide

	Pred.: non-offensive	**Pred.: offensive**
non-offensive	506	114
offensive	87	153

Class 1 (Offensive) Metrics:

$$\text{Recall} = \frac{153}{153 + 87} = \frac{153}{240} \approx 0.638$$

$$\text{Precision} = \frac{153}{153 + 114} = \frac{153}{267} \approx 0.573$$

$$\text{F1} = 2 \cdot \frac{0.573 \times 0.638}{0.573 + 0.638} \approx 0.604$$

Claude Haiku's Confusion Matrix

Predicted	0: non-offensive	1: offensive
0: non-offensive	506	114
1: offensive	87	153

Class 1 (Offensive) Metrics:

$$\text{Recall} = \frac{153}{153 + 114} = \frac{153}{267} \approx 0.573$$

$$\text{Precision} = \frac{153}{153 + 87} = \frac{153}{240} \approx 0.638$$

$$\text{F1} = 2 \cdot \frac{0.638 \times 0.573}{0.638 + 0.573} \approx 0.604$$

Making Mistakes: You Are Not Alone


nature

Explore content | About the journal | Publish with us

nature > articles > article

Article | Published: 01 April 2026

Investigating the replicability of the social and behavioural sciences

Andrew H. Tyner, Anna Lou Abatayo, Mason Daley, Samuel Field, Nicholas Fox, Noah A. Haber,

Figure 2: Replications showed statistically significant results in the original pattern for 151 of 274 claims (55.1% [95% CI 49.2–60.9%]) and for 80.8 of 164 papers (49.3% [95% CI 43.8–54.7%]), weighed for replicating multiple claims per paper (Tyner et al., 2026)

Bias: Good performance $\neq$ same answer - Prevalence

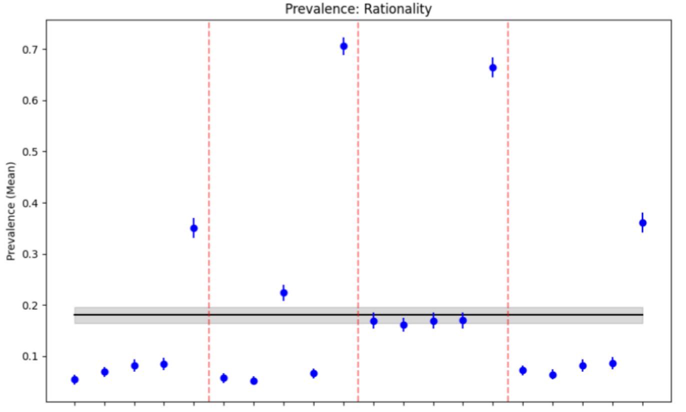


Figure 3: Overview of rationality prevalence in corpus according to different GLLMs relative to manual data reported in Stolwijk et al., 2025

Bias: Good performance $\neq$ same answer - Correlations

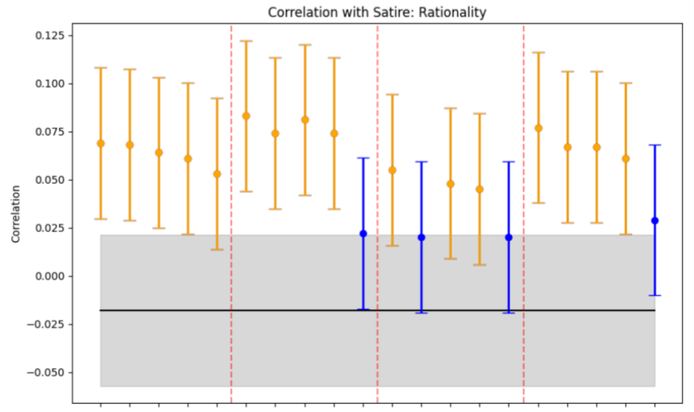


Figure 4: Overview of correlations genre-rationality according to different GLLMs relative to manual data reported in Stolwijk et al., 2025

Good performance ≠ same answer - across models and concepts

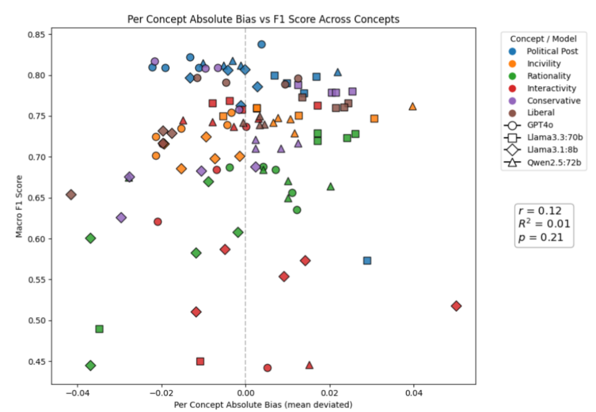


Figure 5: F1 versus correlation bias across concepts and GLLMs, relative to manual data reported in Stolwijk et al., 2025

Remember the gold standard: TweetEval-offensive dataset

We annotated the dataset using the crowdsourcing platform Figure Eight.

We ensured annotation quality by:

- Hiring only experienced annotators
- Using test questions to filter annotators
- Requiring two annotations per tweet
- Requesting a third annotation in case of disagreement
- Using majority vote for final labels

Source: Zampieri et al., 2019

What to do about all this some suggestions

- Being aware of these dangers is half the work
- Always validate a classifier for your task
- Use high-quality manual annotations
- Start your notebook by separating the random test set and keep it isolated until the end.
- Use GitHub or equivalent
- If possible, share data and models
- Annotate and document your code
- Let AI check your code for errors

What to do about all this some suggestions

- Being aware of these dangers is half the work
- Always validate a classifier for your task
- Use high-quality manual annotations
- Start your notebook by separating the random test set and keep it isolated until the end.
- Use GitHub or equivalent
- If possible, share data and models
- Annotate and document your code
- Let AI check your code for errors

What to do about all this some suggestions

- Being aware of these dangers is half the work
- Always validate a classifier for your task
- Use high-quality manual annotations
- Start your notebook by separating the random test set and keep it isolated until the end.
- Use GitHub or equivalent
- If possible, share data and models
- Annotate and document your code
- Let AI check your code for errors

What to do about all this some suggestions

- Being aware of these dangers is half the work
- Always validate a classifier for your task
- Use high-quality manual annotations
- Start your notebook by separating the random test set and keep it isolated until the end.
- Use GitHub or equivalent
- If possible, share data and models
- Annotate and document your code
- Let AI check your code for errors

What to do about all this some suggestions

- Being aware of these dangers is half the work
- Always validate a classifier for your task
- Use high-quality manual annotations
- Start your notebook by separating the random test set and keep it isolated until the end.
- Use GitHub or equivalent
- If possible, share data and models
- Annotate and document your code
- Let AI check your code for errors

What to do about all this some suggestions

- Being aware of these dangers is half the work
- Always validate a classifier for your task
- Use high-quality manual annotations
- Start your notebook by separating the random test set and keep it isolated until the end.
- Use GitHub or equivalent
- If possible, share data and models
- Annotate and document your code
- Let AI check your code for errors

What to do about all this some suggestions

- Being aware of these dangers is half the work
- Always validate a classifier for your task
- Use high-quality manual annotations
- Start your notebook by separating the random test set and keep it isolated until the end.
- Use GitHub or equivalent
- If possible, share data and models
- Annotate and document your code
- Let AI check your code for errors

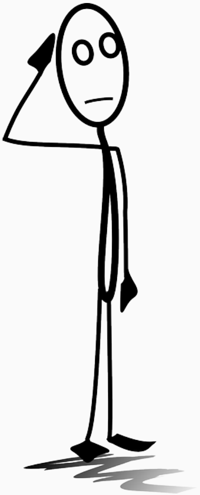
What to do about all this some suggestions

- Being aware of these dangers is half the work
- Always validate a classifier for your task
- Use high-quality manual annotations
- Start your notebook by separating the random test set and keep it isolated until the end.
- Use GitHub or equivalent
- If possible, share data and models
- Annotate and document your code
- Let AI check your code for errors

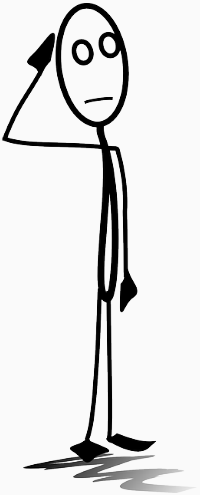
Project day 2

Your takeaway

(short recap of course)



Have your plans about how to and whether to use ML changed?



What are your next steps?

Last part: we help you work on (or discuss) your own projects.

References

Maurer, S., Trilling, et al. (2026). ***Dictionaries, machine learning, transformers or generative LLMs? A framework for deciding which approach to use for computational opinion mining*** **[Working paper, COST Action OPINION (CA21129), WG Tools].**

Stolwijk, S. B., Boukes, M., & Trilling, D. (2025). **Are generative ai text annotations systematically biased?** https://arxiv.org/abs/2512.08404

Stolwijk, S. B., Boukes, M., Yeung, W. N., Liao, Y., Münker, S., Kroon, A. C., & Trilling, D. (2026). **Can we use automated approaches to measure the quality of online political discussion? how to (not) measure interactivity, diversity, rationality, and incivility in online comments to the news.** *Communication Methods and Measures*, *20*(1), 1–25. https://doi.org/10.1080/19312458.2025.2553300

Tyner, A. H., Abatayo, A. L., Daley, M., Field, S., Fox, N., Haber, N. A., Hahn, K. M., Struhl, M. K., Mawhinney, B., Miske, O., et al. (2026). **Investigating the replicability of**

the social and behavioural sciences. *Nature*, *652*(8108), 143–150.

Zampieri, M., Malmasi, S., Nakov, P., Rosenthal, S., Farra, N., & Kumar, R. (2019). **SemEval-2019 task 6: Identifying and categorizing offensive language in social media (OffensEval).** In J. May, E. Shutova, A. Herbelot, X. Zhu, M. Apidianaki, & S. M. Mohammad (Eds.), *Proceedings of the 13th international workshop on semantic evaluation* (pp. 75–86). Association for Computational Linguistics. https://doi.org/10.18653/v1/S19-2010